

MOTOROLA
Semiconductors

BOX 20912 • PHOENIX, ARIZONA 85036

MC6862

(0 to 70°C; L or P Suffix)

MC6862C

(-40 to 85°C; L Suffix only)

Advance Information

2400 bps DIGITAL MODULATOR

The MC6862 is a MOS subsystem designed to be integrated into a wide range of equipment utilizing serial data communication.

The modulator provides the necessary modulation and control functions to implement a serial data communication link over a voice grade channel, utilizing differential phase shift keying (DPSK) at bit rates of 1200 or 2400 bps. Phase options are provided for both the U.S. and international markets. The MC6862 can be implemented into a wide range of data handling systems, including stand-alone modems, data storage devices, remote data communication terminals, and I/O interfaces for counters.

N-channel silicon gate technology permits the MC6862 to operate using a single voltage supply and be fully TTL compatible.

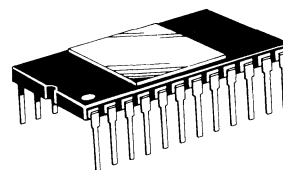
The modulator is compatible with the M6800 microcomputer family, and provides medium-speed data communications capability.

- Clear-to-Send Delay Options
- 511-Bit CCITT Test Pattern
- Terminal Interfaces Are TTL Compatible
- Compatible Functions for 201B/C Data Sets
- CCITT and U.S. Phase Options
- 1200/2400 bps Operation
- Answer Back Tone

MOS

(N-CHANNEL, SILICON-GATE)

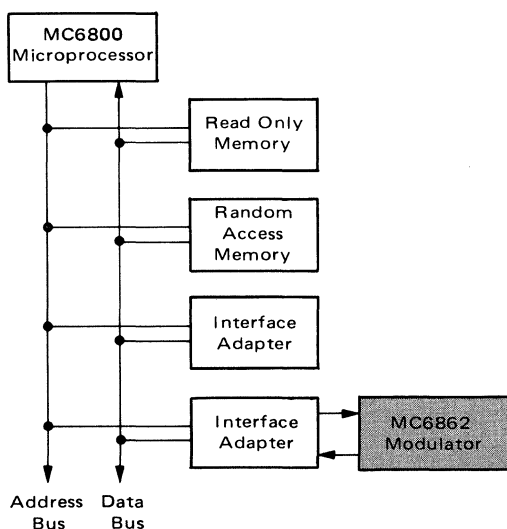
**2400 bps
MODULATOR**



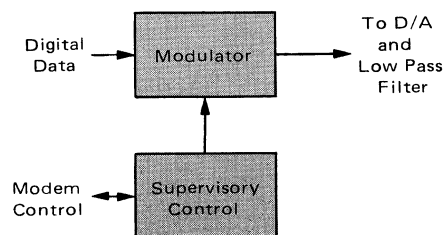
L SUFFIX
CERAMIC PACKAGE
CASE 716

NOT SHOWN: **P SUFFIX**
PLASTIC PACKAGE
CASE 709

M6800 MICROCOMPUTER FAMILY BLOCK DIAGRAM



MC6862 DIGITAL MODULATOR BLOCK DIAGRAM



MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	−0.3 to +7.0	Vdc
Input Voltage	V_{in}	−0.3 to +7.0	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature Range	T_{stg}	−55 to +150	°C
Thermal Resistance	θ_{JA}	82.5	°C/W

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit.

ELECTRICAL CHARACTERISTICS ($V_{DD} = 5.0 \pm 0.25$ Vdc, $V_{SS} = 0$, $T_A = 0$ to 70°C , all outputs loaded as shown in Figure 1 unless otherwise noted.)

Characteristic	Symbol	Min	Typ	Max	Unit
Input High Voltage	V _{IH}	V _{SS} + 2.0	—	V _{DD}	Vdc
Input Low Voltage	V _{IL}	V _{SS}	—	V _{SS} + 0.8	Vdc
Input Current (V _{in} = V _{SS}) CTS1, CTS2, PSS, DRS, $\overline{\text{A}}\text{n}$ $\overline{\text{B}}\text{k}$, and $\overline{\text{T}}\text{x}$ MK RTS and TPE	I _{in}	— —	— —	−0.2 −1.6	mAdc
Input Leakage Current (V _{in} = 5.25 Vdc, V _{DD} = V _{SS})	I _{IL}	—	—	2.5	μAdc
Output High Voltage (I _{OH} = −0.04 mAdc, Load A) (I _{OH} = 0.0 mAdc, Load B)	V _{OH1} V _{OH2}	V _{SS} + 2.4 V _{DD} − 0.5 V	— —	V _{DD} V _{DD}	Vdc
Output Low Voltage (I _{OL} = 1.6 mAdc, Load A)	V _{OL}	V _{SS}	—	V _{SS} + 0.4	Vdc
Input Capacitance (f = 0.1 MHz, T _A = 25°C)	C _{in}	—	5.0	—	pF
V _{DD} Supply Current (All Inputs at V _{SS} except Pin 13 = 57.6 kHz and ALL Outputs Open)	I _{DD}	—	40	60	mAdc
Input Transition Times, All Inputs Except 1.8432 MHz Input (From 10% to 90% points)	t _r t _f	— —	— —	1.0* 1.0*	μs
Input Transition Times, 1.8432 MHz Input (From 0.8 V to 2.0 V)	t _r t _f	— —	— —	40 40	ns
Input Clock Duty Cycle, 1.8432 MHz Input (Measured at 1.5 V level)	D.C.	30	—	70	%
Tx Data Setup Time (Figure 2)	t _S	35	—	—	μs
Tx Data Hold Time (Figure 2)	t _H	35	—	—	μs
Output Transition Times (From 10% to 90% Points)	t _r t _f	— —	— —	5.0 5.0	μs

*Maximum Input Transition Times are $\leq 0.1 \times$ Pulse Width or the specified maximum of $1.0 \mu s$, whichever is smaller.

FIGURE 1 – OUTPUT TEST LOADS

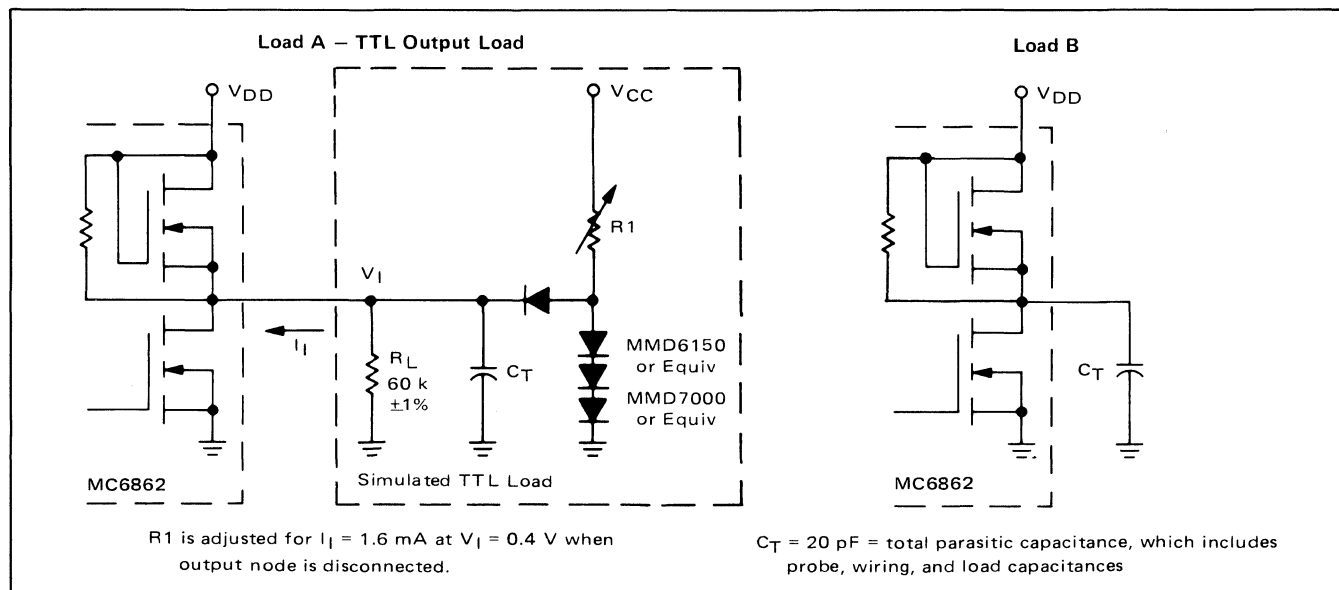
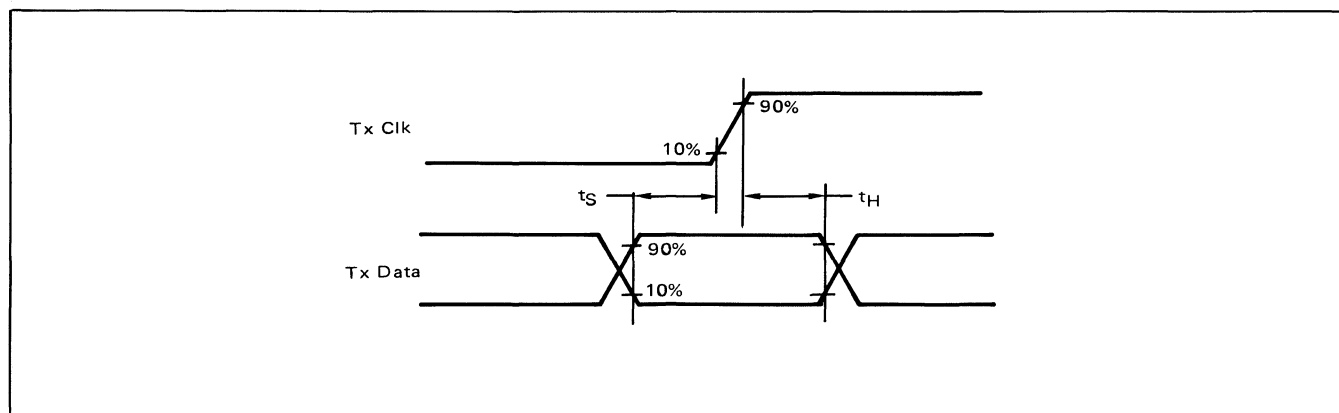
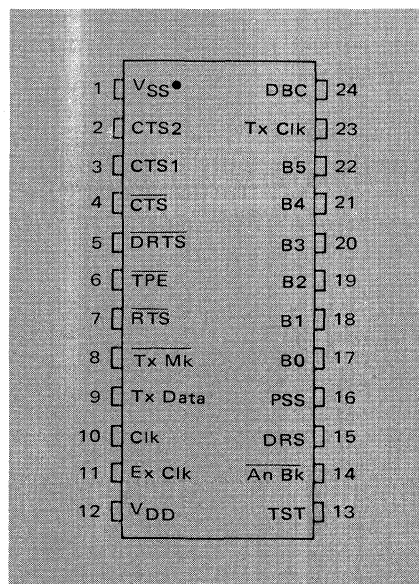


FIGURE 2 – TRANSMIT DATA SETUP AND HOLD TIME



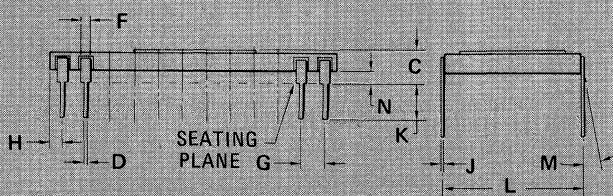
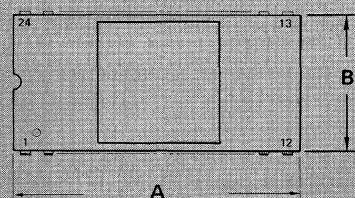
PIN ASSIGNMENT



PACKAGE DIMENSIONS

NOTE:

- LEADS TRUE POSITIONED WITHIN 0.25mm (0.010) DIA (AT SEATING PLANE) AT MAXIMUM MATERIAL CONDITION.



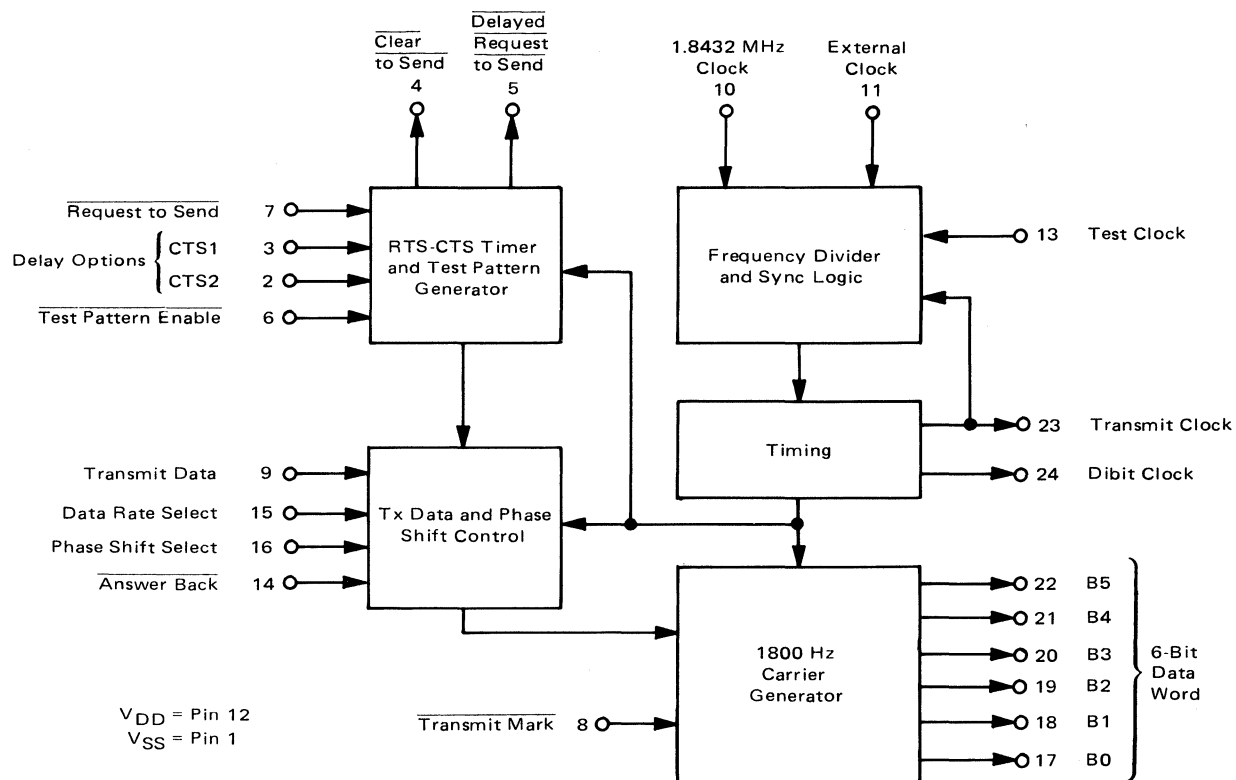
DIM	MILLIMETERS		INCHES	
	MIN	MAX	MIN	MAX
A	29.97	30.99	1.180	1.220
B	14.88	15.62	0.585	0.615
C	3.05	4.19	0.120	0.165
D	0.38	0.53	0.015	0.021
F	0.76	1.40	0.030	0.055
G	2.54 BSC		0.100 BSC	
H	0.76	1.78	0.030	0.070
J	0.20	0.30	0.008	0.012
K	2.54	4.19	0.100	0.165
L	14.88	15.37	0.585	0.605
M	—	10 ⁰	—	10 ⁰
N	0.51	1.52	0.020	0.060

CASE 716-02
(CERAMIC)

SEE PAGE 165 FOR
PLASTIC PACKAGE
DIMENSIONS



BLOCK DIAGRAM



DEVICE OPERATION

GENERAL

Figure 3 shows the modulator and its intraconnections. The data to be transmitted is presented in synchronous serial format to the modulator for conversion to DPSK signals used in transmission. The modulator output is digital; therefore, a D/A converter and a filter transform the signal to an analog form.

The control functions provide four different Clear-to-Send delay options. An Answer Back tone is available for automatic answering applications. The modulator has a built-in 511-bit pseudorandom pattern generator for use in system diagnostic tests.

INPUT/OUTPUT FUNCTIONS

Request to Send (RTS)

The RTS signal from the data terminal controls transmission from the modulator. A low level on RTS activates the modulator data output. A constant mark, for synchronization, is sent during the RTS to CTS delay interval.

Termination of the transmission is accomplished by taking RTS high (see Figures 4 and 5).

Delayed Request to Send (DRTS)

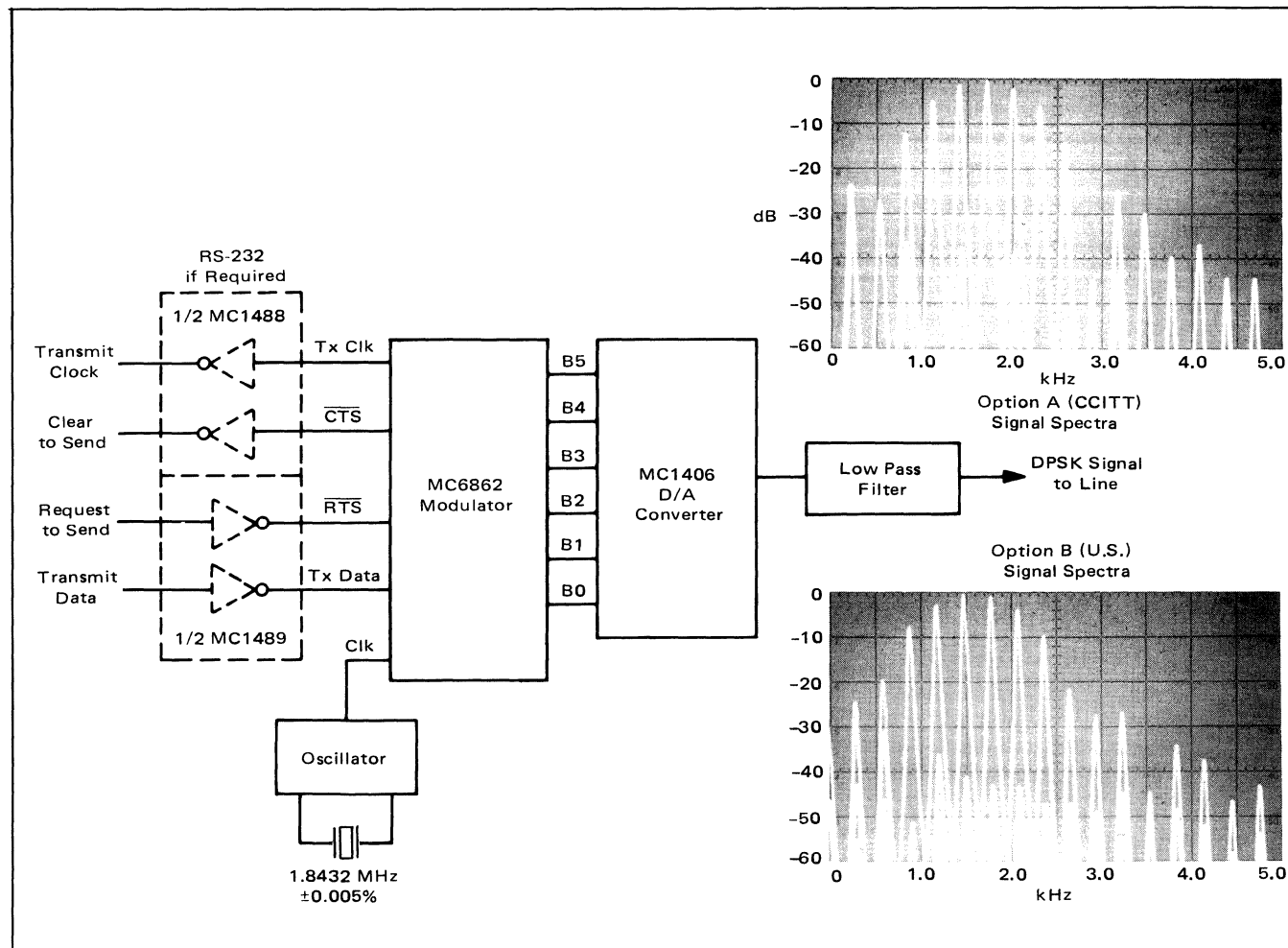
This output can be used to control transmission as specified by the Transmit Mark control input. DRTS follows the negative transition of RTS, and goes negative within the 35 μ s of the negative transition of RTS (Figure 4). The delay from a positive transition of RTS to a positive transition of DRTS is shown in Figure 5. The DRTS delay allows data within the modulator to be transmitted before transmission is inhibited.

Clear to Send (CTS)

CTS follows RTS to both the logic 0 and logic 1 levels. The delay from a negative transition of RTS to a negative CTS transition is selectable by external strapping of CTS1 and CTS2. The delay from a positive transition of RTS to a positive CTS transition is less than 35 μ s.



FIGURE 3 – 2400 BPS MODULATOR INTERFACE



$\overline{\text{CTS}}$ will go low within 35 μs after the positive transition of the Dibit Clock (see Figure 4) except when the no-delay option is selected. For the no-delay option, $\overline{\text{CTS}}$ follows $\overline{\text{RTS}}$ within 35 μs .

RTS-CTS Delay Options (CTS1, CTS2)

The $\overline{\text{RTS}}$ - $\overline{\text{CTS}}$ delays are selectable according to the following strapping options.

RTS-CTS Delay	CTS1	CTS2
0.0 + 0.035 ms, -0.0 ms	0	1
8.55 to 9.35 ms	1	0
24.90 to 26.4 ms	1	1
147.0 to 154.0 ms	0	0

Transmit Mark (Tx Mk)

The Transmit Mark control allows the system designer to select whether the Delayed Request to Send activates

and deactivates the transmission on the modulator chip or off the chip in the output amplifier.

When Tx Mk is high, transmission is controlled on the modulator chip, and occurs from the chip only when $\overline{\text{DRTS}}$ or Answer Back is in the logic 0 state (see Figure 6).

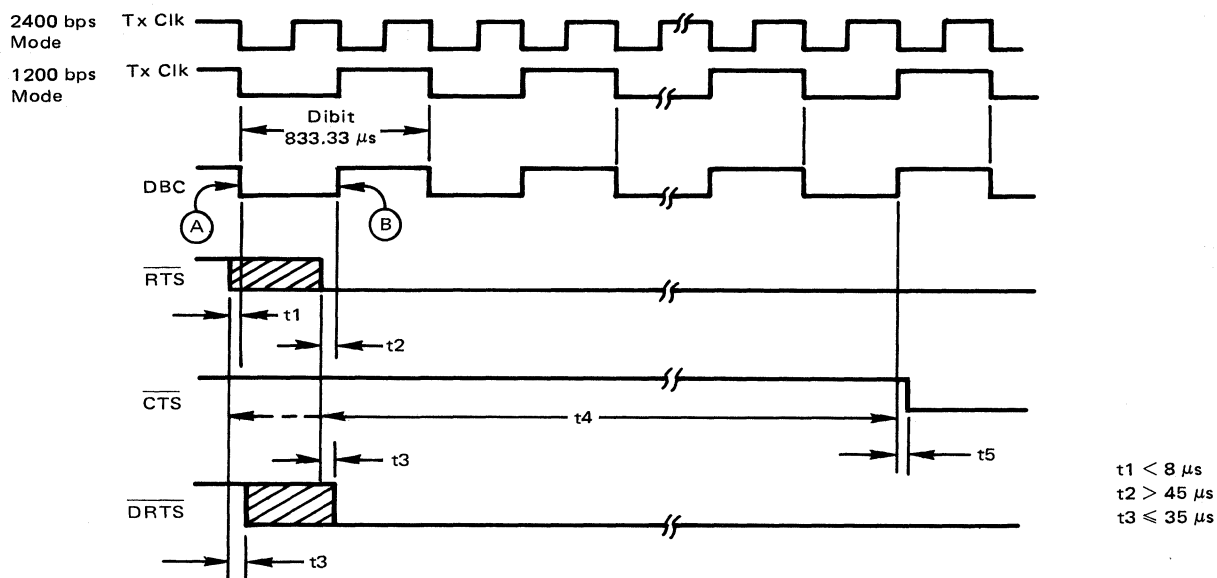
When Tx Mk is low, transmission is controlled off the modulator chip. In this mode, the modulator chip transmits marks at all times except when data or an Answer Back tone is being transmitted (see Figure 6).

Test Pattern Enable (TPE)

A 511-bit test pattern generator is contained on the modulator chip. This pattern is in accord with CCITT specification V52. The pattern can be used to scramble input data, or as a test pattern.

The 511-bit test pattern is activated by applying a logic 0 to TPE. A mark (logic 1) condition on the Transmit Data input with TPE activated (logic 0) causes the test pattern to appear at the data output. A space



FIGURE 4 – $\overline{\text{RTS}}\text{-}\overline{\text{CTS}}$ AND $\overline{\text{RTS}}\text{-}\overline{\text{DRTS}}$ DELAYS

CTS1	CTS2	t4*	t5
0	1	0.0 + 0.035 ms - 0.0 ms	—
1	0	8.55 to 9.35 ms	< 35 μs
1	1	24.90 to 26.4 ms	< 35 μs
0	0	147.0 to 154.0 ms	< 35 μs

*The reference frequency tolerance is not included.

$\overline{\text{RTS}}\text{-}\overline{\text{CTS}}$ delay options are selected by the CTS1 and CTS2 inputs, and are stated as time delay interval t4. An $\overline{\text{RTS}}$ input signal synchronized about point A as shown and going low within the shaded region, will synchronize $\overline{\text{CTS}}$ with the positive transition of DBC (Dibit Clock), and delay t4 is measured with respect to the negative transition of $\overline{\text{RTS}}$.

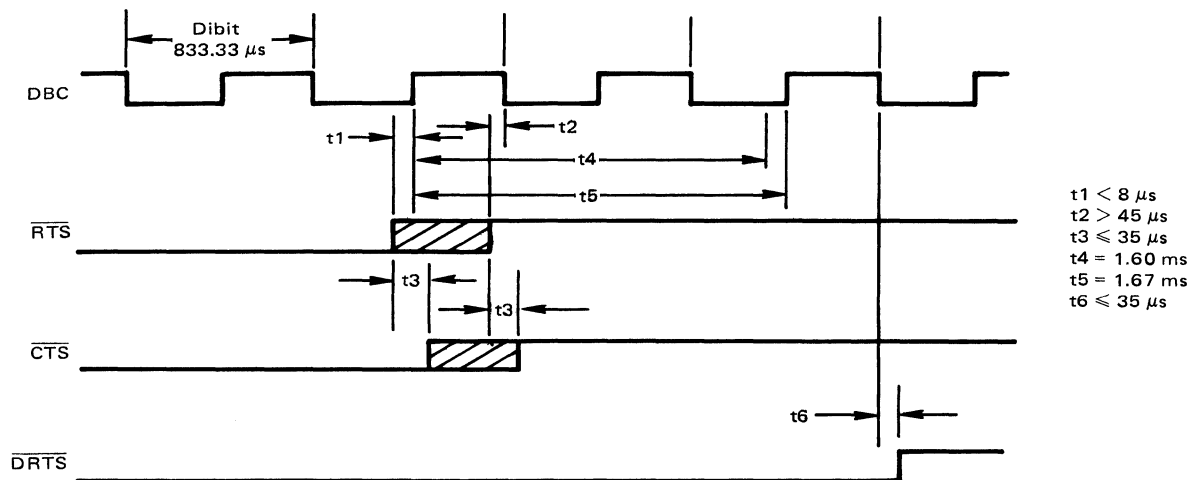
$\overline{\text{RTS}}$ signals going low within the shaded region, but synchronized with the positive transition of DBC (point B), will result in the same $\overline{\text{CTS}}$ delay (t4). For this case the negative transition of $\overline{\text{CTS}}$ is synchronized with the

negative transition of DBC with delay t4 measured with respect to the negative transition of $\overline{\text{RTS}}$.

$\overline{\text{DRTS}}$ will go low within 35 μs of the negative transition of $\overline{\text{RTS}}$ (shown as time t3). With the exception of the no-delay option, $\overline{\text{CTS}}$ will go low within 35 μs (t5) of the positive transition of DBC, following the t4 delay selected. This applies when $\overline{\text{RTS}}$ is synchronized to Point A as shown.

If $\overline{\text{RTS}}$ goes high and remains high $\geq 20 \mu\text{s}$ within time interval t4, a reset of the internal $\overline{\text{RTS}}\text{-}\overline{\text{CTS}}$ timer function will occur. If $\overline{\text{RTS}}$ goes high for less than 20 μs , the circuit may or may not respond to this momentary loss of the $\overline{\text{RTS}}$ signal.



FIGURE 5 – LOSS OF $\overline{\text{RTS}}$ TO $\overline{\text{DRTS}}$ DELAY

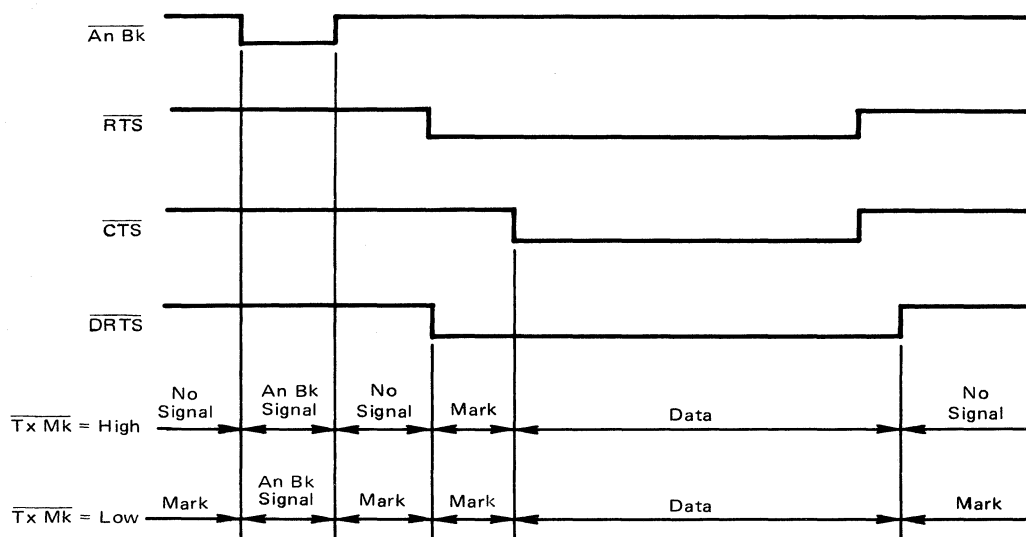
A positive transition of $\overline{\text{RTS}}$ after $\overline{\text{CTS}}$ has become active can result in different functional characteristics of the $\overline{\text{CTS}}$ and $\overline{\text{DRTS}}$ output signals, depending on the time duration that $\overline{\text{RTS}}$ remains inactive.

Under all conditions, $\overline{\text{CTS}}$ will go high within 35 μs (t_3) following a positive transition of $\overline{\text{RTS}}$. If $\overline{\text{RTS}}$ goes high in the shaded region shown (i.e., synchronized to the positive transition of DBC) and remains high beyond the time interval defined as t_5 , then $\overline{\text{DRTS}}$ will go high within 35 μs (t_6) of the next negative transition of DBC.

If $\overline{\text{RTS}}$ were to go low after t_5 , the $\overline{\text{RTS}}$ - $\overline{\text{CTS}}$ delay times given in Figure 4 will result.

If $\overline{\text{RTS}}$ goes high in the shaded region shown, and then returns low within time interval t_4 , the negative transition of $\overline{\text{CTS}}$ will follow within 35 μs , and $\overline{\text{DRTS}}$ will remain in the active or low state. Under these conditions, the normal $\overline{\text{RTS}}$ - $\overline{\text{CTS}}$ delay times are not encountered when $\overline{\text{RTS}}$ is reactivated. If $\overline{\text{RTS}}$ goes low for less than 20 μs , the circuit may or may not respond to this momentary $\overline{\text{RTS}}$ input signal.

FIGURE 6 – TRANSMIT MARK CONTROL



(logic 0) condition on Tx Data with $\overline{TPE}$ activated causes the test pattern data to appear inverted at the data output. Random data applied to Tx Data with $\overline{TPE}$ activated causes the test pattern data to be scrambled (exclusive NORed) with the data, and the result appears at the data output.

The test pattern generator can be enabled only when $\overline{CTS}$ and $\overline{RTS}$ are logic 0. If $\overline{TPE}$ is activated outside this time interval, the previously stated $\overline{RTS-CTS}$ and $\overline{RTS-DRTS}$ delays shown in Figures 4 and 5 are not valid.

Data Rate Select (DRS)

The modulator can transmit at either 2400 bps or 1200 bps. Both data rates utilize an 1800 Hz carrier signal and employ phase shifting at 1200 Hz. The 2400 bps rate is obtained by encoding two bits of data into each phase shift. The 2400 Hz rate is selected by applying a logic 1 to the Data Rate Select lead. The 1200 Hz rate is selected by applying a logic 0 to DRS.

Phase Shift Select (PSS)

Option A (CCITT) or Option B (U.S.) phase shift can be selected for 2400 bps operation. The input data format and phase shift relationship for these two options are as follows:

Data	PSS = 0 Option A	PSS = 1 Option B
00	0°	+45°
01	+90°	+135°
11	+180°	+225°
10	+270°	+315°

For 1200 bps operation, Option C (CCITT) or Option D (U.S.) phase shift can be selected:

Data	PSS = 0 Option C	PSS = 1 Option D
0	+90°	+45°
1	+270°	+225°

Option C is selected by applying a logic 0 to the Phase Shift Select lead when the Data Rate Select lead is strapped for 1200 bps operation (logic 0). Option D is selected by applying a logic 1 to PSS with DRS at logic 0. The phase shifts shown are the difference in phase between the signal at the end of one dibit period and the new signal at the beginning of the next dibit.

Transmit Data (Tx Data)

Transmit Data is the serial binary information presented for DPSK modulation. A high level represents a mark. For timing, see Transmit Clock (Figure 4).

Transmit Clock (Tx Clk)

A 2400/1200 Hz Transmit Clock output is provided for the communication terminal. The Transmit Data signal

is sampled on the positive transition of Transmit Clock. The Transmit Data to Transmit Clock setup and hold time requirements are shown in the Electrical Characteristics table and in Figure 2.

Dibit Clock (DBC)

A 1200 Hz Dibit Clock identifies the modulation timing. This signal goes negative less than 100 μ s prior to the start of dibit modulation.

External Clock (Ex Clk)

A 2400/1200 Hz clock signal applied to the External Clock lead causes Transmit Clock to be synchronized with Ex Clk. This input must have an accuracy within $\pm 0.005\%$.

When no transitions occur on this input, the internal clock provides the 2400/1200 Hz transmit timing signal. Fast synchronization of Tx Clk to Ex Clk is not provided on the chip. *When Ex Clk is not used it should be tied to either the logic 0 or logic 1 state.*

1.8432 MHz (Clk)

This input must be a square wave with rise and fall times of less than 40 ns and a $50 \pm 20\%$ duty cycle. The clock accuracy must be within $\pm 0.005\%$.

Answer Back (An Bk)

A logic 0 level applied to Answer Back causes a 2025 Hz carrier to be generated on the modulator chip instead of a phase shifted 1800 Hz carrier. A logic 1 level applied to An Bk enables the modulator to generate the normal phase shifted 1800 Hz carrier signal, as shown in Figure 6. The time delay from a transition on An Bk to the appropriate signal at the modulator chip output is less than 2 ms.

Activation of An Bk (a logic 0) will disable all other operation modes including the Tx Mk function, and will reset $\overline{CTS}$ to an inactive state along with the $\overline{RTS-CTS}$ internal timer. An Bk should therefore be activated only before initiating $\overline{RTS}$ or after loss of the $\overline{DRTS}$ output signal. The combination of a logic 0 on An Bk with a logic 0 on $\overline{TPE}$ is not used in normal system operation, and hence is used as a reset input during device test.

Digital Output (B0-B5)

These outputs are designed to interface with a six-bit digital-to-analog converter. The resultant signal out of the D/A is the differential phase shift keyed signal quantized at a 14.4 kHz rate. A low pass filter can then be used to smooth the data transitions. B0 is the least significant bit, and the positive level the active state.

Test Clock (TST)

A test signal input is provided to decrease test time of the chip. *In normal operation this input must be strapped low.*

